Reading Environment and History through Tree Rings

日本語 | Choose a language

An introductory course in dendrochronology: eight 90-minute lessons. No previous training in tree-ring measurement or statistics is required.

Lesson 1 requires no preparation. Its videos are for after-class review and independent study. The videos are older slide-reading prototypes and may not match the current materials.

Lessons | Supplementary resources | Offline use

Lessons

Each lesson's details page links to video chapters, exercise data and related resources. PDFs can be opened directly.

Lesson 1: Dating the past and reconstructing environments

Why dates matter and how different dating methods work.

Lesson 1 slides (PDF) · Lesson 1 handout (PDF) · Lesson 1 details and videos

Lesson 2: Tree growth, wood anatomy and tree-ring structure

Wood tissues, ring boundaries and the limits of counting.

Lesson 2 slides (PDF) · Lesson 2 handout (PDF) · Lesson 2 details and videos

Lesson 3: Principles of dendrochronology

Shared growth patterns and crossdating.

Lesson 3 slides (PDF) · Lesson 3 handout (PDF) · Lesson 3 details and videos

Lesson 4: Sampling, measurement and ring-width series

Documenting specimens, imaging, measuring and checking series.

Lesson 4 slides (PDF) · Lesson 4 handout (PDF) · Lesson 4 details and videos

Lesson 5: Dendroarchaeology

From ring formation to felling, construction and human history.

Lesson 5 slides (PDF) · Lesson 5 handout (PDF) · Lesson 5 details and videos

Lesson 6: Dendroecology

Suppression, release and disturbance in forest histories.

Lesson 6 slides (PDF) · Lesson 6 handout (PDF) · Lesson 6 details and videos

Lesson 7: Dendroclimatology

Climate response, calibration, verification and reconstruction limits.

Lesson 7 slides (PDF) · Lesson 7 handout (PDF) · Lesson 7 details and videos

Lesson 8: Research questions linking environment and history

Compare evidence and develop a research plan or literature review.

Lesson 8 slides (PDF) · Lesson 8 handout (PDF) · Lesson 8 details and videos

The course grade is based only on the final report; classroom participation is not graded. See the assessment notice and assignment and rubric. Earlier videos do not define assessment requirements.

Supplementary resources

Use these resources as needed. The mathematics and statistics guide provides optional equations and background.

- Background disciplines (PDF)
- Disturbance and tree responses (PDF)
- Mathematics and statistics (PDF)
- Imaging wood for tree-ring measurement (PDF)
- Finding research papers and using DOIs (PDF)
- Observing wood anatomy in micrographs (PDF)
- Meteorology (PDF)
- Climatology (PDF)
- Paleoclimatology (PDF)
- Tree-ring glossary
- References and databases
- Use and credits

Offline use

Download the older course package (ZIP, including videos)

Extract the ZIP and open START_HERE.html in the en folder. The package is from 30 October 2026 and does not include later updates shown above.